

0.1 级数敛散性判断

0.1.1 估阶法

例题 0.1 判断下述级数收敛性.

1. $\sum_{n=2}^{\infty} \frac{\ln(e^n + n^2)}{\sqrt[4]{n^8 + n^2 + 1} \cdot \ln^2(n+1)}$;
2. $\sum_{n=2}^{\infty} \frac{\sum_{k=1}^n \ln^2 k}{n^p}$;
3. $\sum_{n=1}^{\infty} \frac{1}{n} \int_0^1 \frac{dt}{(1+t^4)^n}$.

证明 [Show/Hide Proof](#)

1. 注意到

$$\frac{\ln(e^n + n^2)}{\sqrt[4]{n^8 + n^2 + 1} \cdot \ln^2(n+1)} = \frac{\ln(e^n) + \ln\left(1 + \frac{n^2}{e^n}\right)}{\sqrt[4]{n^8 + n^2 + 1} \cdot \ln^2(n+1)} \sim \frac{n}{n^2 \cdot \ln^2 n} = \frac{1}{n \ln^2 n}, n \rightarrow \infty.$$

由积分判别法, 我们有

$$\sum \frac{1}{n \ln^2 n} \sim \int_2^{\infty} \frac{1}{x \ln^2 x} dx = \int_{\ln 2}^{\infty} \frac{1}{y^2} dy < \infty,$$

故原级数收敛.

2. 注意到运用 Stolz 定理, 我们有

$$\lim_{n \rightarrow \infty} \frac{1}{n \ln^2 n} \sum_{k=1}^n \ln^2 k = \lim_{n \rightarrow \infty} \frac{\ln^2 n}{n \ln^2 n - (n-1) \ln^2(n-1)} \xrightarrow{\text{拉中保持阶不变}} \lim_{n \rightarrow \infty} \frac{\ln^2 n}{\ln^2 n + 2 \ln n} = 1.$$

于是

$$\sum_{n=2}^{\infty} \frac{\sum_{k=1}^n \ln^2 k}{n^p} \sim \sum_{n=2}^{\infty} \frac{n \ln^2 n}{n^p} = \sum_{n=2}^{\infty} \frac{\ln^2 n}{n^{p-1}}.$$

当 $p > 2$ 时, 取 $\delta > 0$, 使得 $p-1-\delta > 1$. 又 $\lim_{n \rightarrow \infty} \frac{\ln^2 n}{n^\delta} = 0$, 故

$$\sum_{n=2}^{\infty} \frac{\ln^2 n}{n^{p-1}} = \sum_{n=2}^{\infty} \frac{1}{n^{p-1-\delta}} \cdot \frac{\ln^2 n}{n^\delta} < \sum_{n=2}^{\infty} \frac{C}{n^{p-1-\delta}} < +\infty.$$

当 $p \leq 2$ 时, 有

$$\sum_{n=2}^{\infty} \frac{\ln^2 n}{n^{p-1}} > \sum_{n=2}^{\infty} \frac{\ln^2 2}{n^{p-1}} = +\infty.$$

因此原级数收敛等价于 $p > 2$.

3. 由 Laplace 方法, 我们有

$$\int_0^1 \frac{dt}{(1+t^4)^n} = \int_0^1 e^{-n \ln(1+t^4)} dt \sim \int_0^{\infty} e^{-nt^4} dt = \frac{1}{\sqrt[4]{n}} \int_0^{\infty} e^{-x^4} dx, n \rightarrow \infty.$$

现在

$$\frac{1}{n} \int_0^1 \frac{dt}{(1+t^4)^n} \sim \frac{C}{n^{\frac{5}{4}}}, n \rightarrow \infty,$$

故原级数收敛.

□

例题 0.2 设 $a_n = \int_0^{\frac{\pi}{2}} t \left| \frac{\sin^3(nt)}{\sin^3 t} \right| dt, n \in \mathbb{N}$, 判断 $\sum_{n=1}^{\infty} \frac{1}{a_n}$ 收敛性.

笔记 Show/Hide Note

如果需要 a_n 的一个精确的上界, 那么 (就是证法一) 我们可以先待定分段点 θ , 利用不等式 (这个不等式可用数学归纳法证明)

$$|\sin(nt)| \leq n |\sin t|, \forall n \in \mathbb{N}, t \in \mathbb{R},$$

再结合 Jordan 不等式可得

$$\begin{aligned} \int_0^{\frac{\pi}{2}} t \left| \frac{\sin^3(nt)}{\sin^3 t} \right| dt &= \int_0^{\theta} t \left| \frac{\sin^3(nt)}{\sin^3 t} \right| dt + \int_{\theta}^{\frac{\pi}{2}} t \left| \frac{\sin^3(nt)}{\sin^3 t} \right| dt \\ &\leq \int_0^{\theta} \frac{tn^3 \sin^3 t}{\sin^3 t} dt + \int_{\theta}^{\frac{\pi}{2}} \frac{t}{\left(\frac{2}{\pi}t\right)^3} dt \\ &= \frac{\theta^2 n^3}{2} + \frac{\pi^3}{8\theta} - \frac{\pi^2}{4} = g(\theta), \forall n \in \mathbb{N}. \end{aligned}$$

此时再求出 $g(\theta)$ 的最小值点, 就能确定 θ 的取值, 进而得到一个较为精确的上界. 求导易知 $g(\theta) \geq g\left(\frac{\pi}{2n}\right) = \frac{3\pi^2}{8}n$, 即

$$\theta = \frac{\pi}{2n}, a_n \leq \frac{3\pi^2}{8}n, \forall n \in \mathbb{N}.$$

证明 Show/Hide Proof

证法一: 利用不等式

$$|\sin(nt)| \leq n |\sin t|, \forall n \in \mathbb{N}, t \in \mathbb{R},$$

我们有

$$\int_0^{\frac{\pi}{2n}} t \left| \frac{\sin^3(nt)}{\sin^3 t} \right| dt \leq \int_0^{\frac{\pi}{2n}} \frac{tn^3 \sin^3 t}{\sin^3 t} dt = \frac{n^3}{2} \cdot \frac{\pi^2}{4n^2} = \frac{\pi^2}{8}n.$$

利用不等式 $\sin t \geq \frac{2}{\pi}t, \forall t \in \left[0, \frac{\pi}{2}\right]$, 我们有

$$\int_{\frac{\pi}{2n}}^{\frac{\pi}{2}} t \left| \frac{\sin^3(nt)}{\sin^3 t} \right| dt \leq \int_{\frac{\pi}{2n}}^{\frac{\pi}{2}} \frac{t}{\left(\frac{2}{\pi}t\right)^3} dt \leq \frac{\pi^3}{8} \int_{\frac{\pi}{2n}}^{\frac{\pi}{2}} \frac{1}{t^2} dt = \frac{\pi^3}{8} \cdot \frac{2n}{\pi} = \frac{\pi^2}{4}n, \forall n \in \mathbb{N}.$$

现在

$$\int_0^{\frac{\pi}{2}} t \left| \frac{\sin^3(nt)}{\sin^3 t} \right| dt \leq \frac{\pi^2}{8}n + \frac{\pi^2}{4}n = \frac{3\pi^2}{8}n, \forall n \in \mathbb{N},$$

因此

$$\sum_{n=1}^{\infty} \frac{1}{a_n} \geq \frac{8}{3\pi^2} \sum_{n=1}^{\infty} \frac{1}{n} = +\infty.$$

证法二: 注意到 $\lim_{x \rightarrow 0} \frac{\sin^3 x}{x^3} = 1$, 故存在 $c > 0$, 使得

$$|\sin^3 x| \geq cx^3, x \in \left[0, \frac{\pi}{2}\right].$$

从而

$$a_n \leq \frac{1}{c} \int_0^{\frac{\pi}{2}} \frac{|\sin^3(nt)|}{t^2} dt = \frac{n}{c} \int_0^{\frac{n\pi}{2}} \frac{|\sin^3 y|}{y^2} dy \leq \frac{n}{c} \int_0^{\frac{n\pi}{2}} \frac{1}{y^2} dy \leq Kn, \forall n \in \mathbb{N}.$$

其中 K 为某个常数. 于是

$$\sum_{n=1}^{\infty} \frac{1}{a_n} \geq \sum_{n=1}^{\infty} \frac{1}{Kn} = +\infty.$$

□

例题 0.3 对 $x > 0$, 判断级数 $\sum_{n=1}^{\infty} [(2-x)(2-x^{\frac{1}{2}}) \cdots (2-x^{\frac{1}{n}})]$ 收敛性.

证明 Show/Hide Proof

当 $x = 2^m, m \in \mathbb{N}$, 我们知道级数末项在 n 充分大时恒为 0, 因此原本的级数收敛. 下设 $x \neq 2^m, \forall m \in \mathbb{N}$. 此时由

$$\lim_{n \rightarrow \infty} n \left(\frac{(2-x)(2-x^{\frac{1}{2}}) \cdots (2-x^{\frac{1}{n}})}{(2-x)(2-x^{\frac{1}{2}}) \cdots (2-x^{\frac{1}{n+1}})} - 1 \right) = \lim_{n \rightarrow \infty} n \left(\frac{1}{2-x^{\frac{1}{n+1}}} - 1 \right) = \lim_{n \rightarrow \infty} \frac{n(x^{\frac{1}{n+1}} - 1)}{2-x^{\frac{1}{n+1}}} = \lim_{n \rightarrow \infty} n \frac{\ln x}{n+1} = \ln x,$$

以及 Raabe 判别法, 我们就有 $x > e$ 时级数收敛, $x < e$ 时级数发散. 当 $x = e$ 时, 计算 Taylor 展开式得

$$\ln \frac{1}{2-e^{\frac{1}{n+1}}} = \frac{1}{n} + \frac{1}{12n^4} + o\left(\frac{1}{n^4}\right), n \rightarrow \infty,$$

然后

$$\lim_{n \rightarrow \infty} \ln n \left[n \ln \frac{1}{2-e^{\frac{1}{n+1}}} - 1 \right] = 0.$$

运用较为广泛的判别法 (极限版 2), 我们知道原级数发散.

□

例题 0.4 判断级数 $\sum_{n=1}^{\infty} x^{\sin 1 + \sin \frac{1}{2} + \cdots + \sin \frac{1}{n}}, x \in (0, 1)$ 收敛性.

证明 Show/Hide Proof

注意到

$$\begin{aligned} \lim_{n \rightarrow \infty} n \left(\frac{x^{\sin 1 + \sin \frac{1}{2} + \cdots + \sin \frac{1}{n}}}{x^{\sin 1 + \sin \frac{1}{2} + \cdots + \sin \frac{1}{n+1}}} - 1 \right) &= \lim_{n \rightarrow \infty} n \left(\frac{1}{x^{\sin \frac{1}{n+1}}} - 1 \right) \\ &= \lim_{n \rightarrow \infty} n \left(e^{-\sin \frac{1}{n+1} \cdot \ln x} - 1 \right) = -\ln x \lim_{n \rightarrow \infty} n \sin \frac{1}{n+1} = -\ln x. \end{aligned}$$

由 Raabe 判别法, 我们有 $0 < x < \frac{1}{e}$ 时原级数收敛, $\frac{1}{e} < x < 1$ 时原级数发散. 当 $x = \frac{1}{e}$, 由较为广泛的判别法 (极限版 2) 和

$$\begin{aligned} \lim_{n \rightarrow \infty} \ln n \left(n \ln \frac{x^{\sin 1 + \sin \frac{1}{2} + \cdots + \sin \frac{1}{n}}}{x^{\sin 1 + \sin \frac{1}{2} + \cdots + \sin \frac{1}{n+1}}} - 1 \right) &= \lim_{n \rightarrow \infty} \ln n \left(n \ln \frac{1}{x^{\sin \frac{1}{n+1}}} - 1 \right) \\ &= \lim_{n \rightarrow \infty} \ln n \left(n \ln e^{-\sin \frac{1}{n+1} \ln x} - 1 \right) = \lim_{n \rightarrow \infty} \ln n \left(n \left[e^{-\sin \frac{1}{n+1} \ln x} - 1 + O\left((e^{-\sin \frac{1}{n+1} \ln x} - 1)^2\right) \right] - 1 \right) \\ &= \lim_{n \rightarrow \infty} \ln n \left(n \left[-\sin \frac{1}{n+1} \ln x + O\left(\left(-\sin \frac{1}{n+1} \ln x\right)^2\right) + O\left(\frac{1}{n^2}\right) \right] - 1 \right) \\ &= \lim_{n \rightarrow \infty} \ln n \left(\left[n \sin \frac{1}{n+1} + O\left(\frac{1}{n}\right) \right] - 1 \right) = \lim_{n \rightarrow \infty} \left[\ln n \cdot O\left(\frac{1}{n}\right) \right] = 0, \end{aligned}$$

我们有原级数在 $x = \frac{1}{e}$ 发散.

□

0.1.2 带对数换底法

例题 0.5 判断下列级数收敛性.

1. $\sum_{n=2}^{\infty} \frac{1}{r^{\ln n}} (r > 0);$
2. $\sum_{n=2}^{\infty} \frac{1}{\ln^{\ln n} n};$
3. $\sum_{n=3}^{\infty} \frac{1}{(\ln \ln n)^{\ln n}};$
4. $\sum_{n=3}^{\infty} \frac{1}{(\ln n)^{\ln \ln n}};$

$$5. \sum_{n=2}^{\infty} \frac{1}{n^{1+\frac{1}{\sqrt{\ln n}}}};$$

$$6. \sum_{n=3}^{\infty} \frac{1}{n^{1+\frac{1}{\ln \ln n}} \ln n}$$

证明 Show/Hide Proof

1. 注意到

$$\sum_{n=2}^{\infty} \frac{1}{r^{\ln n}} = \sum_{n=2}^{\infty} \frac{1}{e^{\ln n \cdot \ln r}} = \sum_{n=2}^{\infty} \frac{1}{n^{\ln r}},$$

我们有原级数收敛等价于 $r > e$.

2. 注意到

$$\sum_{n=100}^{\infty} \frac{1}{\ln^{\ln n} n} = \sum_{n=100}^{\infty} \frac{1}{e^{\ln n \cdot \ln \ln n}} = \sum_{n=100}^{\infty} \frac{1}{n^{\ln \ln n}} \leq \sum_{n=100}^{\infty} \frac{1}{n^{\ln \ln 100}} < \infty,$$

我们有原级数收敛.

3. 注意到

$$\sum_{n=[e^{e^e}]+1}^{\infty} \frac{1}{(\ln \ln n)^{\ln n}} = \sum_{n=[e^{e^e}]+1}^{\infty} \frac{1}{e^{\ln n \cdot \ln \ln \ln n}} \leq \sum_{n=[e^{e^e}]+1}^{\infty} \frac{1}{e^{\ln n \cdot \ln \ln \ln(e^{e^e}+1)}} = \sum_{n=[e^{e^e}]+1}^{\infty} \frac{1}{n^{\ln \ln \ln(e^{e^e}+1)}} < \infty,$$

我们有原级数收敛.

4. 由积分判别法, 我们有

$$\sum_{n=3}^{\infty} \frac{1}{(\ln n)^{\ln \ln n}} = \sum_{n=3}^{\infty} \frac{1}{e^{\ln^2 \ln n}} \sim \int_3^{\infty} \frac{1}{e^{\ln^2 \ln x}} dx \stackrel{x=e^y}{=} \int_{\ln 3}^{\infty} e^{y - \ln^2 y} dy = \infty,$$

故原级数发散.

5. 首先

$$\sum_{n=2}^{\infty} \frac{1}{n^{1+\frac{1}{\sqrt{\ln n}}}} = \sum_{n=2}^{\infty} \frac{1}{ne^{\sqrt{\ln n}}},$$

于是我们结合

$$\sum_{n=2}^{\infty} \frac{1}{ne^{\sqrt{\ln n}}} \leq \sum_{n=2}^{\infty} \int_{n-1}^n \frac{1}{xe^{\sqrt{\ln x}}} dx = \int_1^{\infty} \frac{1}{xe^{\sqrt{\ln x}}} dx = \int_0^{\infty} \frac{1}{e^{\sqrt{t}}} dt < \infty$$

知原级数收敛.

6. 当 n 充分大有

$$\frac{1}{n^{1+\frac{1}{\ln \ln n}} \ln n} = \frac{1}{e^{(1+\frac{1}{\ln \ln n}) \ln n} \ln n} = \frac{1}{n \ln ne^{\frac{\ln n}{\ln \ln n}}} \leq \frac{1}{n \ln ne^{\frac{(\ln \ln n)^2}{\ln \ln n}}} = \frac{1}{n \ln ne^{\ln \ln n}} = \frac{1}{n \ln^2 n}.$$

由积分判别法知

$$\sum \frac{1}{n \ln^2 n} \sim \int_2^{\infty} \frac{1}{x \ln^2 x} dx = \int_{\ln 2}^{\infty} \frac{1}{y^2} dy < \infty,$$

因此 $\sum_{n=3}^{\infty} \frac{1}{n^{1+\frac{1}{\ln \ln n}} \ln n}$ 收敛.

□

0.1.3 Taylor 公式法

例题 0.6 判断 $\sum_{n=1}^{\infty} \frac{(-1)^n}{(-1)^{n-1} + \sqrt[n]{n}}$ 收敛性.

 **笔记** Show/Hide Note

此类问题可采用 Taylor 公式的 peano 余项方法, 主要要展开到余项的级数绝对收敛. 注意积累绝对 ± 绝对 = 绝对, 绝对 ± 条件 = 条件的结论.

证明 Show/Hide Proof

注意到

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{(-1)^{n-1} + \sqrt[3]{n}} = \sum_{n=1}^{\infty} \frac{\frac{(-1)^n}{\sqrt[3]{n}}}{1 - \frac{(-1)^n}{\sqrt[3]{n}}} = \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{\sqrt[3]{n}} + \frac{1}{n^{\frac{2}{3}}} + \frac{(-1)^n}{n} + O\left(\frac{1}{n^{\frac{4}{3}}}\right) \right).$$

于是由 $\sum_{n=1}^{\infty} \left(\frac{(-1)^n}{\sqrt[3]{n}} + \frac{(-1)^n}{n} + O\left(\frac{1}{n^{\frac{4}{3}}}\right) \right)$ 收敛, $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{2}{3}}}$ 发散知原级数发散.

□

0.1.4 分组判别法和利用不等式判别

例题 0.7 设 $a_n > 0$, $\sum_{n=1}^{\infty} a_n$ 收敛, 证明级数 $\sum_{n=1}^{\infty} a_n^{\frac{n}{n+1}}$ 收敛.

 **笔记** Show/Hide Note待定常数 c , 记

$$M \triangleq \left\{ n \in \mathbb{N} : a_n \leq c a_n^{\frac{n}{n+1}} \right\}, N \triangleq \left\{ n \in \mathbb{N} : a_n > c a_n^{\frac{n}{n+1}} \right\}.$$

则对 $\forall n \in \mathbb{N}$, 有

$$a_n > c a_n^{\frac{n}{n+1}} \Rightarrow \frac{1}{c} > a_n^{\frac{1}{n+1}} \Rightarrow a_n < \left(\frac{1}{c} \right)^{n+1}.$$

现在我们想要利用几何级数将原级数进行分组, 故只需任取 $c > 1$ 即可, 不妨取 $c = 2$, 就有下述证明.

证明 Show/Hide Proof

记

$$M \triangleq \left\{ n \in \mathbb{N} : a_n \leq \frac{1}{2} a_n^{\frac{n}{n+1}} \right\}, N \triangleq \left\{ n \in \mathbb{N} : a_n > \frac{1}{2} a_n^{\frac{n}{n+1}} \right\}.$$

现在

$$a_n^{\frac{1}{n+1}} \leq \frac{1}{2} \Rightarrow a_n \leq \left(\frac{1}{2} \right)^{n+1} \Rightarrow a_n^{\frac{n}{n+1}} \leq \left(\frac{1}{2} \right)^n, \forall n \in M,$$

我们有

$$\sum_{n=1}^{\infty} a_n^{\frac{n}{n+1}} = \sum_{n \in M} a_n^{\frac{n}{n+1}} + \sum_{n \in N} a_n^{\frac{n}{n+1}} \leq \sum_{n \in M} \left(\frac{1}{2} \right)^n + 2 \sum_{n \in N} a_n < \infty,$$

这就完成了证明.

□

例题 0.8 设 f 是 $[1, +\infty)$ 上定义的正连续函数, 且无穷积分 $\int_1^{\infty} f(x) dx$ 收敛.

(1) 问: 是否必有 $\lim_{x \rightarrow +\infty} f(x) = 0$? 说明理由.

(2) 无穷积分 $\int_1^{\infty} f(x)^{1-\frac{1}{x}} dx$ 是否收敛?

解 Show/Hide Solution

(1) 不是. 比如

$$f(x) = \begin{cases} 1 - n^2|x - n|, & n - \frac{1}{n^2} \leq x \leq n + \frac{1}{n^2}, \\ 0, & \text{其它 } x \end{cases}$$

满足

$$\int_1^{\infty} f(x) dx = \frac{1}{2} + \sum_{n=2}^{\infty} \frac{1}{n^2} < \infty,$$

但由 $\lim_{n \rightarrow \infty} f(n) = 1$ 知 $\lim_{x \rightarrow +\infty} f(x) = 0$ 不成立.

(2) 证法一: 令

$$A \triangleq \left\{x \geq 1 \mid f^{1-\frac{1}{x}}(x) \leq 2f(x)\right\}, \quad B \triangleq \left\{x \geq 1 \mid f^{1-\frac{1}{x}}(x) > 2f(x)\right\} = \left\{x \geq 1 \mid f(x) < \frac{1}{2^x}\right\}.$$

则结合条件可知

$$\begin{aligned} \int_1^\infty f^{1-\frac{1}{x}}(x) dx &= \int_A f^{1-\frac{1}{x}}(x) dx + \int_B f^{1-\frac{1}{x}}(x) dx \leq 2 \int_A f(x) dx + \int_B \left(\frac{1}{2^x}\right)^{1-\frac{1}{x}} dx \\ &= 2 \int_A f(x) dx + \int_B \frac{1}{2^{x-1}} dx \leq 2 \int_A f(x) dx + \int_1^\infty \frac{1}{2^{x-1}} dx < \infty. \end{aligned}$$

证法二: 由 Young 不等式可得

$$\begin{aligned} \int_1^\infty f^{1-\frac{1}{x}}(x) dx &= 2 \int_1^\infty \frac{1}{2} f^{\frac{x-1}{x}}(x) dx \leq 2 \int_1^\infty \left[\frac{1}{x2^x} + \frac{x-1}{x} \left(f^{\frac{x-1}{x}}(x)\right)^{\frac{x}{x-1}} \right] dx \\ &= 2 \int_1^\infty \frac{1}{x2^x} dx + 2 \int_1^\infty \frac{x-1}{x} f(x) dx \leq 2 \int_1^\infty \frac{1}{2^x} dx + 2 \int_1^\infty f(x) dx < +\infty. \end{aligned}$$

□

例题 0.9 设 $\alpha > 0$, 正项级数 $\sum_{k=1}^\infty a_k$ 收敛. 证明:

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n (1 + a_k^{1-k^{-\alpha}})$$

存在.

证明 Show/Hide Proof

证法一 (分组判别法): 记

$$S = \left\{k \in \mathbb{N} : a_k^{1-k^{-\alpha}} > ea_k\right\} = \left\{k \in \mathbb{N} : a_k < \frac{1}{e^{k^\alpha}}\right\},$$

则

$$\begin{aligned} \sum_{k=1}^n \ln(1 + a_k^{1-k^{-\alpha}}) &= \sum_{k \in S} \ln(1 + a_k^{1-k^{-\alpha}}) + \sum_{k \in \mathbb{N} \setminus S} \ln(1 + a_k^{1-k^{-\alpha}}) \\ &\leq \sum_{k \in S} a_k^{1-k^{-\alpha}} + \sum_{k \in \mathbb{N} \setminus S} a_k^{1-k^{-\alpha}} \leq \sum_{k \in S} \left(\frac{1}{e^{k^\alpha}}\right)^{1-k^{-\alpha}} + \sum_{k \in \mathbb{N} \setminus S} ea_k \\ &\leq \sum_{k \in S} \left(\frac{1}{e}\right)^{k^\alpha-1} + e \sum_{k=1}^n a_k = e \sum_{k=1}^n \left(\frac{1}{e}\right)^{k^\alpha} + e \sum_{k=1}^n a_k. \end{aligned}$$

令 $n \rightarrow +\infty$, 得

$$\sum_{k=1}^\infty \ln(1 + a_k^{1-k^{-\alpha}}) \leq e \sum_{k=1}^\infty \left(\frac{1}{e}\right)^{k^\alpha} + e \sum_{k=1}^\infty a_k.$$

由积分判别法知, 级数 $\sum_{k=1}^\infty \left(\frac{1}{e}\right)^{k^\alpha}$ 与反常积分 $\int_1^\infty e^{-x^\alpha} dx$ 同敛散. 注意到

$$\int_1^\infty e^{-x^\alpha} dx \stackrel{x=t^{\frac{1}{\alpha}}}{=} \frac{1}{\alpha} \int_1^\infty t^{\frac{1}{\alpha}-1} e^{-t} dt = \frac{1}{\alpha} \Gamma\left(\frac{1}{\alpha}\right) < +\infty,$$

故 $\sum_{k=1}^\infty \left(\frac{1}{e}\right)^{k^\alpha} < +\infty$, 而由题设知 $\sum_{k=1}^\infty a_k < +\infty$, 因此 $\sum_{k=1}^\infty \ln(1 + a_k^{1-k^{-\alpha}})$ 收敛. 由此可知

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n (1 + a_k^{1-k^{-\alpha}}) = \lim_{n \rightarrow \infty} e^{\sum_{k=1}^n \ln(1 + a_k^{1-k^{-\alpha}})} = e^{\sum_{k=1}^\infty \ln(1 + a_k^{1-k^{-\alpha}})} < +\infty.$$

证法二 (Young 不等式): 利用 Young 不等式

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

取

$$a = a_k^{1-k^{-\alpha}}, \quad b = \frac{1}{2}, \quad p = \frac{1}{1-k^{-\alpha}}, \quad q = k^{\alpha},$$

则对 $\forall k \in \mathbb{N}$, 有

$$\begin{aligned} \frac{a_k^{1-k^{-\alpha}}}{2} &\leq (1-k^{-\alpha})a_k + \frac{1}{k^{\alpha}} \left(\frac{1}{2}\right)^{k^{\alpha}} \leq a_k + \frac{1}{k^{\alpha}} \left(\frac{1}{2}\right)^{k^{\alpha}} \\ &\Rightarrow a_k^{1-k^{-\alpha}} \leq 2a_k + \frac{2}{k^{\alpha}} \left(\frac{1}{2}\right)^{k^{\alpha}}. \end{aligned}$$

从而

$$\sum_{k=1}^{\infty} a_k^{1-k^{-\alpha}} \leq 2 \sum_{k=1}^{\infty} a_k + 2 \sum_{k=1}^{\infty} \frac{1}{k^{\alpha}} \left(\frac{1}{2}\right)^{k^{\alpha}}. \quad (1)$$

由 $\lim_{k \rightarrow \infty} \frac{k^{\alpha}}{\ln k} = +\infty$ 知, 存在 $K \in \mathbb{N}$, 使得

$$\frac{k^{\alpha}}{\ln k} > \frac{2}{\ln 2} \iff k^{\alpha} \ln 2 > 2 \ln k \iff \left(\frac{1}{2}\right)^{k^{\alpha}} = e^{-k^{\alpha} \ln 2} < e^{-2 \ln k} = \frac{1}{k^2}, \quad \forall k \geq K.$$

于是

$$\sum_{k=K}^{\infty} \frac{1}{k^{\alpha}} \left(\frac{1}{2}\right)^{k^{\alpha}} \leq \sum_{k=K}^{\infty} \frac{1}{k^{2+\alpha}} < +\infty.$$

故 $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}} \left(\frac{1}{2}\right)^{k^{\alpha}}$ 收敛, 又由题设知 $\sum_{k=1}^{\infty} a_k < +\infty$, 故由(1)式知 $\sum_{k=1}^{\infty} a_k^{1-k^{-\alpha}}$ 也收敛. □

例题 0.10 若 $\sum_{n=1}^{\infty} a_n$ 收敛, 其中 $a_n > 0, 0 < \alpha, \beta < 1, \alpha + \beta > 1$, 证明级数 $\sum_{n=1}^{\infty} \frac{a_n^{\alpha}}{n^{\beta}}$ 收敛.

证明 [Show/Hide Proof](#)

对 $\forall n \in \mathbb{N}$, 由 Young 不等式知

$$\frac{a_n^{\alpha}}{n^{\beta}} \leq \frac{a_n^{\alpha \cdot \frac{1}{\alpha}}}{\frac{1}{\alpha}} + \frac{n^{\beta \cdot \frac{1}{1-\alpha}}}{\frac{1}{1-\alpha}} = \alpha a_n + \frac{1-\alpha}{n^{\frac{\beta}{1-\alpha}}}.$$

又由 $0 < \alpha, \beta < 1$ 且 $\alpha + \beta > 1$ 知 $\frac{\beta}{1-\alpha} > 1$, 故

$$\sum_{n=1}^{\infty} \frac{a_n^{\alpha}}{n^{\beta}} \leq \alpha \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} \frac{1-\alpha}{n^{\frac{\beta}{1-\alpha}}} < +\infty. \quad \square$$

例题 0.11

- (1) 证明 $\sum_{n=1}^{\infty} \frac{(-1)^{\lfloor \sqrt{n} \rfloor}}{n}$ 收敛.
- (2) 讨论 $\sum_{n=1}^{\infty} \frac{(-1)^{\sqrt{n}}}{n^p}$ ($p > 0$) 的敛散性.

注 对 $\forall n \in \mathbb{N}$, 都存在 $t \in \mathbb{N}$, 使得

$$t \leq \sqrt{n} < t+1 \iff t^2 \leq n \leq (t+1)^2 - 1.$$

因此

$$(-1)^{\lfloor \sqrt{k} \rfloor} = (-1)^t, \quad \forall k \in [t^2, (t+1)^2 - 1].$$

并且还有

$$\ln(1+x) \leq x \Rightarrow \ln\left(1 - \frac{1}{k}\right) \leq -\frac{1}{k} \Rightarrow \frac{1}{k} \geq \ln\left(\frac{k}{k-1}\right).$$

注 虽然(1)的证法二中的 O 估计只对 n 充分大时成立, 但实际上级数的前有限项的和一定是有限数, 因此讨论级

数的敛散性时, 可以只讨论从 n 的充分大的一项开始的级数. 上述证明中省略了这步, 总体问题不大.

证明 Show/Hide Proof

(1) **证法一:** 由级数加括号的理理解得

$$\sum_{n=1}^{\infty} \frac{(-1)^{\lfloor \sqrt{n} \rfloor}}{n} = \sum_{t=1}^{\infty} \sum_{k=t^2}^{(t+1)^2-1} \frac{(-1)^{\lfloor \sqrt{k} \rfloor}}{k} = \sum_{t=1}^{\infty} (-1)^t \sum_{k=t^2}^{(t+1)^2-1} \frac{1}{k}.$$

我们来证明上式右边级数符合莱布尼兹判别法.

首先

$$0 \leq \lim_{t \rightarrow \infty} \sum_{k=t^2}^{(t+1)^2-1} \frac{1}{k} \leq \lim_{t \rightarrow \infty} \sum_{k=t^2}^{(t+1)^2-1} \int_{k-1}^k \frac{1}{x} dx = \lim_{t \rightarrow \infty} \sum_{k=t^2}^{(t+1)^2-1} \ln \frac{k}{k-1} = \lim_{t \rightarrow \infty} \ln \frac{(t+1)^2-1}{t^2-1} = 0,$$

即

$$\lim_{t \rightarrow \infty} \sum_{k=t^2}^{(t+1)^2-1} \frac{1}{k} = 0.$$

然后对 $t \in \mathbb{N}, t \geq 3$, 我们有

$$\begin{aligned} \sum_{k=t^2}^{(t+1)^2-1} \frac{1}{k} - \sum_{k=(t+1)^2}^{(t+2)^2-1} \frac{1}{k} &= \sum_{k=0}^{2t} \frac{1}{k+t^2} - \sum_{k=0}^{2t+2} \frac{1}{k+t^2+2t+1} \\ &= \sum_{k=0}^{2t} \left(\frac{1}{k+t^2} - \frac{1}{k+t^2+2t+1} \right) - \frac{1}{t^2+4t+2} - \frac{1}{t^2+4t+3} \\ &= \sum_{k=0}^{2t} \frac{2t+1}{(k+t^2)(k+t^2+2t+1)} - \frac{2t^2+8t+5}{(t^2+4t+2)(t^2+4t+3)} \\ &\geq \frac{(2t+1)^2}{(2t+t^2)(2t+t^2+2t+1)} - \frac{2t^2+8t+5}{(t^2+4t+2)(t^2+4t+3)} \\ &\geq \frac{(2t+1)^2}{(t^2+2t+2)(t^2+4t+3)} - \frac{2t^2+8t+5}{(t^2+4t+2)(t^2+4t+3)} \\ &= \frac{2t^2-4t-4}{(t^2+4t+2)(t^2+4t+3)} \geq \frac{6t-4t-4}{(t^2+4t+2)(t^2+4t+3)} > 0, \end{aligned}$$

这就证明了 $\sum_{k=t^2}^{(t+1)^2-1} \frac{1}{k}$ 单调递减! 现在由莱布尼兹判别法得 $\sum_{n=1}^{\infty} \frac{(-1)^{\lfloor \sqrt{n} \rfloor}}{n}$ 收敛.

证法二: 由级数加括号的理理解得

$$\sum_{n=1}^{\infty} \frac{(-1)^{\lfloor \sqrt{n} \rfloor}}{n} = \sum_{t=1}^{\infty} \sum_{k=t^2}^{(t+1)^2-1} \frac{(-1)^{\lfloor \sqrt{k} \rfloor}}{k} = \sum_{t=1}^{\infty} (-1)^t \sum_{k=t^2}^{(t+1)^2-1} \frac{1}{k}.$$

我们用估阶方法进行证明. 由例题????可知

$$\sum_{k=1}^n \frac{1}{k} = \ln n + \gamma + O\left(\frac{1}{n}\right).$$

从而

$$\sum_{k=1}^{t^2-1} \frac{1}{k} = \ln(t^2-1) + \gamma + O\left(\frac{1}{t^2}\right),$$

$$\sum_{k=1}^{(t+1)^2-1} \frac{1}{k} = \ln[(t+1)^2-1] + \gamma + O\left(\frac{1}{t^2}\right).$$

于是

$$\sum_{k=t^2}^{(t+1)^2-1} \frac{1}{k} = \ln \frac{t^2+2t}{t^2-1} + O\left(\frac{1}{t^2}\right) = \ln\left(1 + \frac{2t+1}{t^2-1}\right) + O\left(\frac{1}{t^2}\right).$$

进而

$$\sum_{t=1}^{\infty} (-1)^t \sum_{k=t^2}^{(t+1)^2-1} \frac{1}{k} = \sum_{t=1}^{\infty} (-1)^t \left[\ln \left(1 + \frac{2t+1}{t^2-1} \right) + O \left(\frac{1}{t^2} \right) \right].$$

显然 $\ln \left(1 + \frac{2t+1}{t^2-1} \right) + O \left(\frac{1}{t^2} \right)$ 单调递减趋于 0, 故由莱布尼兹判别法可知原级数收敛.

(2) 注意到

$$\sum_{n=1}^{\infty} \frac{(-1)^{\sqrt{n}}}{n^p} = \sum_{k=1}^{\infty} (-1)^k \sum_{n=k^2}^{(k+1)^2-1} \frac{1}{n^p}.$$

对 $\forall k \geq 2$, 有

$$\begin{aligned} \sum_{n=k^2}^{(k+1)^2-1} \frac{1}{n^p} &\leq \sum_{n=k^2}^{(k+1)^2} \frac{1}{n^p} = \frac{1}{k^{2p}} + \sum_{n=k^2+1}^{(k+1)^2} \frac{1}{n^p} \\ &\leq \frac{1}{k^{2p}} + \sum_{n=k^2+1}^{(k+1)^2} \int_{n-1}^n \frac{1}{x^p} dx = \frac{1}{k^{2p}} + \int_{k^2}^{(k+1)^2} \frac{1}{x^p} dx \\ &= \frac{1}{k^{2p}} + \frac{(k+1)^{2-2p} - k^{2-2p}}{1-p}, \\ \sum_{n=k^2}^{(k+1)^2-1} \frac{1}{n^p} &\geq \sum_{n=k^2}^{(k+1)^2-1} \int_n^{n+1} \frac{1}{x^p} dx = \int_{k^2}^{(k+1)^2} \frac{1}{x^p} dx = \frac{(k+1)^{2-2p} - k^{2-2p}}{1-p}. \end{aligned}$$

于是

$$\begin{aligned} \overline{\lim}_{k \rightarrow +\infty} \frac{1}{2k^{1-2p}} \sum_{n=k^2}^{(k+1)^2-1} \frac{1}{n^p} &\leq \overline{\lim}_{k \rightarrow +\infty} \frac{(k+1)^{2-2p} - k^{2-2p}}{2(1-p)k^{1-2p}} = 1, \\ \underline{\lim}_{k \rightarrow +\infty} \frac{1}{2k^{1-2p}} \sum_{n=k^2}^{(k+1)^2-1} \frac{1}{n^p} &\geq \underline{\lim}_{k \rightarrow +\infty} \left[\frac{1}{2k} + \frac{(k+1)^{2-2p} - k^{2-2p}}{2(1-p)k^{1-2p}} \right] = 1. \end{aligned}$$

故

$$\sum_{n=k^2}^{(k+1)^2-1} \frac{1}{n^p} \sim 2k^{1-2p}, \quad k \rightarrow +\infty.$$

因此

$$\sum_{n=1}^{\infty} \frac{(-1)^{\sqrt{n}}}{n^p} \sim 2 \sum_{k=1}^{\infty} (-1)^k k^{1-2p}.$$

当 $p \leq \frac{1}{2}$ 时, 上述级数通项 $(-1)^k k^{1-2p}$ 发散, 故此时 $\sum_{n=1}^{\infty} \frac{(-1)^{\sqrt{n}}}{n^p}$ 发散;

当 $p > \frac{1}{2}$ 时, 由莱布尼兹判别法知, $\sum_{n=1}^{\infty} \frac{(-1)^{\sqrt{n}}}{n^p}$ 收敛.

□

例题 0.12 设 $a_n > 0$, $\sum_{n=1}^{\infty} a_n$ 收敛, 证明级数 $\sum_{n=1}^{\infty} \frac{a_n^{\frac{1}{p}}}{n^s}$, $s > 1 - \frac{1}{p}$, $p > 1$ 收敛.

注 本题也可用分组判别法进行证明.

证明 Show/Hide Proof

利用 Young 不等式, 我们有

$$\sum_{n=1}^{\infty} \frac{a_n^{\frac{1}{p}}}{n^s} \leq \sum_{n=1}^{\infty} \left(\frac{a_n}{p} + \frac{1}{qn^{sq}} \right) < \infty, \quad sq = \frac{s}{1 - \frac{1}{p}} > 1,$$

这就完成了证明.

□

0.1.5 拟合法和积分判别法

例题 0.13 设 $n \in \mathbb{N}, a_n > 1$ 且单调递增, $\alpha > 0$.

1. 证明: $\sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1} a_n^\alpha} < +\infty$
2. 证明: $\sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1} (\ln a_n)^{1+\alpha}} < +\infty$

证明 Show/Hide Proof

由 $\{a_n\}$ 递增知 $\lim_{n \rightarrow \infty} a_n \in [1, +\infty]$, 进而 $\lim_{n \rightarrow \infty} \frac{1}{a_n^\alpha} \in [0, 1)$.

(1) 注意到

$$\sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1}^\alpha} = \sum_{n=1}^{+\infty} \int_{a_n}^{a_{n+1}} \frac{dx}{a_{n+1}^\alpha} \leq \sum_{n=1}^{+\infty} \int_{a_n}^{a_{n+1}} \frac{dx}{x^{\alpha+1}} = \frac{1}{\alpha} \left(\frac{1}{a_1^\alpha} - \lim_{n \rightarrow \infty} \frac{1}{a_{n+1}^\alpha} \right) < +\infty,$$

故

$$\begin{aligned} \sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1} a_n^\alpha} &= \sum_{n=1}^{+\infty} \left(\frac{a_{n+1} - a_n}{a_{n+1} a_n^\alpha} - \frac{a_{n+1} - a_n}{a_{n+1}^\alpha} \right) + \sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1}^\alpha} \\ &= \sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1}} \left(\frac{1}{a_n^\alpha} - \frac{1}{a_{n+1}^\alpha} \right) + \sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1}^\alpha} \\ &< \sum_{n=1}^{+\infty} \left(\frac{1}{a_n^\alpha} - \frac{1}{a_{n+1}^\alpha} \right) + \sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1}^\alpha} \\ &= \frac{1}{a_1^\alpha} - \lim_{n \rightarrow \infty} \frac{1}{a_{n+1}^\alpha} + \sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1}^\alpha} < +\infty. \end{aligned}$$

(2) 注意到

$$\begin{aligned} \sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1} (\ln a_{n+1})^{1+\alpha}} &= \sum_{n=1}^{+\infty} \int_{a_n}^{a_{n+1}} \frac{dx}{a_{n+1} (\ln a_{n+1})^{\alpha+1}} \leq \sum_{n=1}^{+\infty} \int_{a_n}^{a_{n+1}} \frac{dx}{x \ln^{\alpha+1} x} \\ &\leq \int_{a_1}^{+\infty} \frac{dx}{x \ln^{\alpha+1} x} = \frac{1}{\alpha \ln^\alpha a_1} < +\infty, \end{aligned}$$

又

$$\sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1} (\ln a_n)^{1+\alpha}} = \sum_{n=1}^{+\infty} \left[\frac{a_{n+1} - a_n}{a_{n+1} (\ln a_n)^{1+\alpha}} - \frac{a_{n+1} - a_n}{a_{n+1} (\ln a_{n+1})^{1+\alpha}} \right] + \sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1} (\ln a_{n+1})^{1+\alpha}},$$

故只需证

$$\sum_{n=1}^{+\infty} \left[\frac{a_{n+1} - a_n}{a_{n+1} (\ln a_n)^{1+\alpha}} - \frac{a_{n+1} - a_n}{a_{n+1} (\ln a_{n+1})^{1+\alpha}} \right] < +\infty.$$

现在

$$\begin{aligned} \sum_{n=1}^{+\infty} \left[\frac{a_{n+1} - a_n}{a_{n+1} (\ln a_n)^{1+\alpha}} - \frac{a_{n+1} - a_n}{a_{n+1} (\ln a_{n+1})^{1+\alpha}} \right] &= \sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1}} \left[\frac{1}{(\ln a_n)^{1+\alpha}} - \frac{1}{(\ln a_{n+1})^{1+\alpha}} \right] \\ &\leq \sum_{n=1}^{+\infty} \frac{a_{n+1} - a_n}{a_{n+1}} \left[\frac{1}{(\ln a_n)^{1+\alpha}} - \frac{1}{(\ln a_{n+1})^{1+\alpha}} \right] \\ &< \sum_{n=1}^{+\infty} \left[\frac{1}{(\ln a_n)^{1+\alpha}} - \frac{1}{(\ln a_{n+1})^{1+\alpha}} \right] \\ &= \frac{1}{(\ln a_1)^{1+\alpha}} - \lim_{n \rightarrow \infty} \frac{1}{(\ln a_n)^{1+\alpha}} < +\infty. \end{aligned}$$

故结论得证.

□

0.1.6 杂题

例题 0.14 证明: $\sum_{n=1}^{\infty} \frac{\sin n}{n}, \sum_{n=1}^{\infty} \frac{\cos n}{n}$ 条件收敛.

证明 Show/Hide Proof

利用积化和差公式得

$$\begin{aligned} \left| \sum_{j=1}^n \sin j \right| &= \left| \sum_{j=1}^n \frac{\sin j \cdot \sin \frac{1}{2}}{\sin \frac{1}{2}} \right| = \left| \sum_{j=1}^n \frac{\cos(j - \frac{1}{2}) - \cos(j + \frac{1}{2})}{2 \sin \frac{1}{2}} \right| \\ &= \left| \frac{\cos \frac{1}{2} - \cos(n + \frac{1}{2})}{2 \sin \frac{1}{2}} \right| = \frac{1}{2} \left| \sin n - \cot \frac{1}{2} \cos n + \cot \frac{1}{2} \right| \\ &\leq \frac{2}{2 \sin \frac{1}{2}} = \frac{1}{\sin \frac{1}{2}}, \\ \left| \sum_{j=1}^n \cos j \right| &= \left| \sum_{j=1}^n \frac{\cos j \cdot \sin \frac{1}{2}}{\sin \frac{1}{2}} \right| = \left| \sum_{j=1}^n \frac{\sin(j + \frac{1}{2}) - \sin(j - \frac{1}{2})}{2 \sin \frac{1}{2}} \right| \\ &= \left| \frac{\sin(n + \frac{1}{2}) - \sin \frac{1}{2}}{2 \sin \frac{1}{2}} \right| = \frac{1}{2} \left| \cos n + \cot \frac{1}{2} \sin n - 1 \right| \\ &\leq \frac{2}{2 \sin \frac{1}{2}} = \frac{1}{\sin \frac{1}{2}}, \end{aligned}$$

都是有界的. 又 $\frac{1}{n}$ 递减到 0, 我们由 A-D 判别法得 $\sum_{n=1}^{\infty} \frac{\sin n}{n}, \sum_{n=1}^{\infty} \frac{\cos n}{n}$ 收敛.

又

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{|\sin n|}{n} &\geq \sum_{n=1}^{\infty} \frac{|\sin n|^2}{n} = \sum_{n=1}^{\infty} \frac{1 - \cos(2n)}{2n}; \\ \sum_{n=1}^{\infty} \frac{|\cos n|}{n} &\geq \sum_{n=1}^{\infty} \frac{|\cos n|^2}{n} = \sum_{n=1}^{\infty} \frac{1 + \cos(2n)}{2n}. \end{aligned}$$

故 $\sum_{n=1}^{\infty} \frac{\sin n}{n}, \sum_{n=1}^{\infty} \frac{\cos n}{n}$ 条件收敛.

□

例题 0.15 设 $\{a_n\}$ 为实数列, $S_n = a_1 + a_2 + \cdots + a_n$, $\sigma_n = \frac{1}{n+1}(S_1 + S_2 + \cdots + S_n)$. 已知级数 $\sum_{n=1}^{\infty} |S_n - \sigma_n|^2$ 收敛,

求证: $\sum_{n=1}^{\infty} a_n$ 收敛.

证明 Show/Hide Proof

注意到对 $\forall n \geq 2$, 有

$$\begin{aligned} \sigma_n - \sigma_{n-1} &= \frac{\sum_{k=1}^n S_k}{n+1} - \frac{\sum_{k=1}^{n-1} S_k}{n} = \frac{n \sum_{k=1}^n S_k - (n+1) \sum_{k=1}^{n-1} S_k}{n(n+1)} \\ &= \frac{nS_n - \sum_{k=1}^{n-1} S_k}{n(n+1)} = \frac{(n+1)S_n - \sum_{k=1}^n S_k}{n(n+1)} = \frac{S_n - \sigma_n}{n}. \end{aligned}$$

再由 Cauchy-Schwarz 不等式和题设可得

$$\lim_{n \rightarrow \infty} (\sigma_n - \sigma_1) = \sum_{n=2}^{\infty} (\sigma_n - \sigma_{n-1}) \leq \sum_{n=2}^{\infty} |\sigma_n - \sigma_{n-1}| = \sum_{n=2}^{\infty} \frac{|S_n - \sigma_n|}{n} \leq \sqrt{\sum_{n=2}^{\infty} |S_n - \sigma_n|^2} \cdot \sqrt{\sum_{n=2}^{\infty} \frac{1}{n^2}} < +\infty.$$

故可设 $\lim_{n \rightarrow \infty} \sigma_n \triangleq L < +\infty$, 且

$$\lim_{n \rightarrow \infty} |\sigma_n - \sigma_{n-1}| = \lim_{n \rightarrow \infty} \frac{|S_n - \sigma_n|}{n} = 0 \implies \sum_{n=1}^{\infty} a_n = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sigma_n = L.$$

□